CHAHYON KU

PhD Student
Robotics Department
University of Michigan

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SUMMARY

I am a PhD student at University of Michigan advised by Assistant Professor Bernadette Bucher, currently working on Robot Learning for Mobile Manipulation.

SKILLS

Languages: Python, C/C++

Technologies: PyTorch, JAX, CUDA, ROS, Bullet

EDUCATION

9/2024 - 5/2028 **Doctor of Philosophy in Robotics** — **University of Michigan**
- Advisor: Bernadette Bucher
- Research: Robot Learning for Mobile Manipulation

9/2022 - 5/2024 **Master of Science in Robotics** — **University of Minnesota**
- Advisor: Karthik Desingh
- Research: Imitation Learning for Manipulation, Object-centric Representations

9/2016 - 8/2022 **Bachelor of Science in Computer Science** — **University of Washington**
- Research: Object-centric Representations, Instance Segmentation
- Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing

PUBLICATIONS

Chahyon Ku, Carl Winge, Ryan Diaz, Wentao Yuan, and Karthik Desingh. Evaluating Robustness of Visual Representations for Object Assembly Task Requiring Spatio-Geometrical Reasoning. *IEEE International Conference on Robotics and Automation (ICRA), 2024.*

RESEARCH EXPERIENCE

Workshop Paper **HELIOS: Hierarchical Exploration for Language-grounded Interaction in Open Scenes**
- Presented as 10-min Spotlight at CoRL 2025 Learning Effective Abstractions for Planning Workshop.
- Worked on a hierarchical exploration agent that performs open vocabulary mobile manipulation by utilizing VLM value maps for global planning and uncertainty-aware gaussian splats for local planning.

Conference Pub. **Evaluating Robustness of Visual Representations for Object Assembly Task Requiring Spatio-Geometrical Reasoning** — **https://sites.google.com/view/geometric-peg-in-hole**
- Presented as Oral at ICRA 2024
- Presented as 8-min Spotlight at CoRL 2023 Pretraining for Robot Learning Workshop
- Proposed and implemented a novel dual-arm robotic manipulation task involving the assembly of parts with a specific geometric relationship, modeling real-world tasks such as capping a bottle
- Evaluated the performance of pretrained vision encoders through imitation learning in simulation and real

Undergrad. Res. **Evaluating SORNet on a Geometric and Spatial Reasoning Dataset** — **chahyon-ku.github.io/sornet-geospa**
- Undergraduate research at UW Robotics and State Estimation Lab (PI: Postdoc Scholar Karthik Desingh)
- Extended *SORNet: Spatial Object-Centric Representations for Sequential Manipulation* (CoRL 2021) to predict the geometric and spatial relations as predicates from RGB images
- Generated simulated images of elementary shapes in various configurations (supported, contained, etc.)
- Performed comparative analysis on sensitivity to unseen object atrributes and relations

Undergrad. Res. **University of Washington-Amazon Robot Manipulation Project**
- Undergraduate research at UW Robotics and State Estimation Lab (Mentor: PhD Candidate Yi Li)
- Worked on building a system of UR16 and RGBD camera to pick objects from Amazon pods
- Generated simulated RGBD images of randomized bins using the Google Scanned Objects (NVISII)
- Implemented, trained, and evaluated a U-net-based baseline for instance segmentation of products

PROFESSIONAL EXPERIENCE

9/2025 – 12/2025 **Intern, Research** — **NASA Jet Propulsion Laboratory**
- Conduct research in indoor-outdoor navigation from natural language instructions.
- Produce a real-robot demo of a rover with a 7-DoF arm navigating office / lab campus environments.

6/2025 – 9/2025 **Intern, Perception** **Waymo**

- Reimplemented and Extended a Recent State-of-the-art Method for 3D Object Detection from Camera and LIDAR Observations.

6/2023 – 9/2023 **Intern, Perception** **Zoox**

- Developed computer vision models for improving autonomous driving behavior around pedestrians
- Identified the problem and mined 1 million relevant samples using proprietary C++/Python codebase
- Designed and conducted experiments to present findings in documents and presentations
- Communicated with various teams on the AI stack for feedback and smoother integration onto the vehicle

TEACHING EXPERIENCE

9/2023 – 5/2024 **Graduate Teaching Assistant** **University of Minnesota**

- CSCI 5551 Introduction to Intelligent Robotics
- Adapted the kineval-stencil framework by Professor Chad Jenkins to fit contents of the course
- Implemented solutions and improved the autograder with detailed error messages.
- Created a new homework assignment combining previous assignments in kinematics, path planning, and state machines to simulate a mobile pick-and-place task.

4/2022 – 6/2022 **Undergraduate Teaching Assistant** **University of Washington**

- CSE 473 Artificial Intelligence
- Created and graded problem sets on search, markov decision processes, and reinforcement learning.